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Predicting Cancer Therapy Response Through Patient-Specific Genome-Scale Metabolic Models

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Abstract

Motivation: Approximately half of all cancer patients fail to respond to first-line therapy, yet treatment is chosen by cancer type rather than individual tumour genome. Metabolism is the proximal output of gene expression, so patient-specific GSMMs encode the metabolite exchange rates of each tumour using the reaction fluxes and may reveal flux-level signatures that stratify therapy responders from non-responders for each therapy type.

Results: We integrated 11,505 TCGA RNA-seq profiles into the Recon3D human metabolic model with 10,600 reactions using the GIMME algorithm, yielding an $11{,}505 \times 10{,}600$ reaction-flux matrix. Differential flux analysis (Mann-Whitney U for large cohorts ($n > 500$), permutation test for small cohorts ($n < 500$) were combined with Fisher's exact test and Benjamini-Hochberg correction at FDR < 0.05) recovered 47 significant reactions for Chemotherapy and 95 for Radiation Therapy, tracing back to 53 and 81 unique genes respectively. LightGBM classifiers with Optuna hyperparameter tuning achieved out-of-fold AUCs of 0.785 (Chemotherapy) and 0.818 (Radiation Therapy) using binary reaction features combined with clinical covariates. SHAP analysis confirms that cancer site and disease stage dominate prediction, with metabolic features supplying interpretable secondary signal mechanistically consistent with nucleotide-pool maintenance and altered carbon-source utilisation as resistance mechanisms.

Keywords: cancer therapy response prediction; genome-scale metabolic models; flux balance analysis; TCGA; Recon3D; GIMME; metabolic biomarkers; LightGBM; SHAP; differential flux analysis

Abbreviations: GSMMs, Genome-Scale Metabolic Models; TCGA, The Cancer Genome Atlas; RNA-seq, RNA sequencing; FBA, Flux Balance Analysis; GIMME, Gene Inactivity Moderated by Metabolism and Expression; FDR, False Discovery Rate; AUC, Area Under the Receiver Operating Characteristic Curve; SHAP, SHapley Additive exPlanations.

1 Introduction

1.1 The Clinical Problem

Approximately half of cancer patients fail to respond to first-line therapy [7]. Tumours of identical histological type or primary cancer type can exhibit different metabolic phenotypes, and current treatment selection is based primarily on cancer type rather than individual molecular biology or to be more specific, the tumour genome. Each failed treatment course costs months of disease progression and substantial economic burden in toxic, ineffective therapy. A principled framework that ranks therapy options before treatment begins would substantially reduce this clinical burden.

1.2 Why Metabolism?

Metabolism is the proximal output of gene expression: every signalling, growth, and stress-response programme ultimately routes through metabolic flux. Two patients with identical oncogenic mutations can exhibit divergent responses to chemotherapy when their metabolic contexts differ [2]. GSMMs encode this context in a stoichiometric matrix $\mathbf{S}$ whose entries describe how each reaction consumes or produces each metabolite; Flux Balance Analysis (FBA) then solves $\max_{\mathbf{v}} \mathbf{c}^\top \mathbf{v}$ subject to $\mathbf{Sv} = \mathbf{0}$ and flux bounds. Patient-specific GSMMs built by pruning the reference network with transcriptomic evidence yield a mechanistic readout of which biochemistry is active in each individual tumour.

1.3 Related Work and Motivation

Lewis and Kemp [7] demonstrated that GSMMs built from ∼915 TCGA cancer patients multi-omcis data and analysed with gradient-boosted classifiers achieved AUC ≈ 0.85 for radiation response prediction using metabolite production rates as features. The present work scales this approach by a factor of $12.5\times$ in patient count to 11,505 samples, extends it to multi-therapy stratification across six treatment classes, and provides reaction-level rather than metabolite-level biomarkers. This work also compares the biomarker reactions obtained between therapies and provides a classifier that will ultimately suggest a patient which therapy to choose, which was not done earlier and that is the motivation for this study. Earlier metabolic-modelling efforts [3, 5] established Recon3D as the reference human metabolic model and the COBRA Toolbox as the algorithmic foundation upon

which this work builds.

1.4 Central Hypothesis

If a tumour's metabolic state predicts how it responds to therapy, then patient's tumour-specific GSMMs constructed from baseline RNA-seq data should identify reactions whose flux distinguishes responders from non-responders, and these biomarker reactions should support classification of therapy outcome.

1.5 Objectives

1. Construct 11,505 patient's tumour-specific GSMMs from TCGA pan-cancer bulk RNA-seq data and perform FBA.
2. Do Quality Control (QC) on all the models.
3. Perform therapy-stratified differential flux analysis to identify biomarker reactions of response.
4. Backtrack significant reactions to their encoding genes and substrate metabolites.
5. Train LightGBM ML classifiers for each therapy type with SHAP-based explanations.
6. Compare the biomarker reactions obtained between each therapy type, along with its interpretation.

2 Materials and Methods

2.1 Data Sources

Three data streams were integrated (Table 1): (i) TCGA RNA-seq — STAR-augmented gene counts (TPM, unstranded) for 11,505 patients across 33 cancer types, downloaded from the GDC Data Portal; (ii) TCGA Clinical — patient-level demographics and treatment outcomes annotated by RECIST criteria (3,721 patient-therapy rows covering 6 therapy classes). Multiple rows per patient where a patient received multiple therapies.
(iii) Recon3D — the reference human metabolic model with 10,600 reactions, 5,835 metabolites and 2,248 metabolic genes spanning 9 subcellular compartments [3]. Gene-symbol mapping employed the MyGene.info batch API (1,675 genes resolved from Entrez IDs to HGNC symbols).

Table 1. Summary of datasets used in this study.

Source	Count
TCGA RNA-seq	11,505 files
TCGA Clinical	3,721 patient-therapy rows
Recon3D (XML/MAT)	10,600 reactions

2.2 Sensitivity Labelling

Therapy outcome was classified using RECIST criteria: *Sensitive* (Complete Response, Partial Response, No Measurable Disease) versus *Resistant* (Stable Disease, Persistent Disease, Progressive Disease, No Response). Records labelled Unknown, Not Reported, Treatment Ongoing, or missing were excluded. The 49 raw treatment strings were collapsed into six therapy groups; only Chemotherapy (n = 1,719) and Radiation Therapy (n = 1,595) were significant enough for differential analysis (Table 1).

Table 1: Therapy-stratified cohort sizes.

Therapy	Sens.	Res.	Total	Significance
Chemotherapy	1127	592	1719	High
Radiation	1177	418	1595	High
Hormone	84	32	116	Low
Surgery	98	10	108	Low
Immunotherapy	48	58	106	Low
Targeted	23	54	77	Low
Total	2557	1164	3721	

2.3 GIMME Algorithm

Each RNA-seq profile was processed as follows: STAR summary rows were dropped, duplicates were resolved by max-TPM, and values were $\log_e$-transformed. The GIMME algorithm [2] was then applied to build a patient-specific reduction of Recon3D:

Phase 1 — Biomass maximisation

$$\max \quad \mathbf{c}^\top \mathbf{v}$$

subject to

$$\mathbf{S}\mathbf{v} = 0$$

$$\mathbf{v}^{\text{lb}} \leq \mathbf{v} \leq \mathbf{v}^{\text{ub}}$$

to obtain

$$v^*_{\max}$$

Phase 2 — Consistency-penalised LP

$$\min \quad \sum_j w_j |v_j|$$

subject to

$$\mathbf{S}\mathbf{v} = 0$$

$$\mathbf{c}^\top \mathbf{v} \geq \theta \cdot v^*_{\max} \qquad (\theta = 0.85)$$

$$w_j = \max(0,\ \varepsilon - r_j)$$

where:

$$r_j = \text{GPR expression score}$$

$$\varepsilon = \ln(10)$$

GPR aggregation rules:

$$\text{AND}: \quad r_j = \min(\text{expr})$$

$$\text{OR}: \quad r_j = \max(\text{expr})$$

Gene-controlled reactions with $w_j > 0$ and near-zero flux ($|v_j| < 10^{-6}$) are subsequently pruned as they unnecessarily force the model to rewire other reactions to keep these reactions which don't have significant flux rate. Exchange and demand reactions (no GPR) are always retained. GPR rules use AND (enzyme complex →

minimum expression) and OR (isozymes → maximum expression). On average ∼7,500 reactions and ∼1,500 genes were retained per model. GLPK was used as the LP solver for all GIMME steps (Gurobi licence was unavailable at that stage); subsequent FBA and biomarker steps used Gurobi after the WLS licence was activated. Both solvers guarantee LP optimality.

2.4 Flux Balance Analysis

FBA was run on each of the 11,505 patient models, optimising for biomass production. Fluxes were aligned to the canonical 10,600-reaction Recon3D ordering and merged into a single HDF5 matrix of shape 11,505 × 10,600 (`fba_matrix.h5`). **Silent reactions** were identified, which are active reactions in the Recon 3D model but inactive in the patient specific model. On average, around 4000 - 6000 reactions were silent per model and this is because, the patient specific models are site specific but Recon 3D is for the entire body. A QC report tracked objective value, dead-end metabolites, structural reactions and gap-filling requirements per model. On average, there were 7900 reactions, 1557 genes, 663 (FBA solution), 1430 dead-end metabolites and 4662 structural reactions per model. Thermokernel[8] algorithm was not available in COBRApy[4], and couldn't run the MATLAB version of the code in the actinium server due to licensing issues of MATLAB. Tried running the loopless FVA algorithm in COBRApy, but since it is a MILP-based problem, it was extremely slow for large-scale models. Because of that, FVA was not performed completely and the flux ranges for individual reactions could not be obtained to identify blocked reactions for subsequent gap filling. Hence, gap filling was also not performed. However, the thermokernel algorithm was tested on 10 patient models in my local system, where it successfully generated around 700–800 reactions per model for 6 patients, but failed for the remaining cases and was unable to fully remove TICs in all models.

2.5 Differential Flux Analysis

A reaction can differ between Sensitive and Resistant cohorts in two biologically distinct ways: (1) both groups use it but at different magnitudes; (2) it is active in one group and silent in the other. A single test cannot capture both modes simultaneously.

2.5.1 Magnitude test

For large cohorts ($N > 500$), the non-parametric Mann-Whitney U test is applied to raw flux values as the distribution is non-normal and the spread of samples in cohorts is skewed towards the class of sensitive. For small cohorts ($N < 500$), a vectorised permutation test with 10,000 label shuffles is used. The observed mean-difference between the two groups is compared to the empirical null:

$$p_{\text{perm}} = \frac{\#\{k : |\bar{v}_S^{(k)} - \bar{v}_R^{(k)}| \geq |\bar{v}_S^{\text{obs}} - \bar{v}_R^{\text{obs}}|\} + 1}{n_{\text{perm}} + 1}$$

The vectorised implementation generates all 10,000 permutation index matrices in NumPy at once, achieving a ∼100× speedup over a naive Python loop.

2.5.2 Presence/absence test

Fisher's exact test is applied to the 2×2 contingency table of {present, absent}×{Sensitive, Resistant}, where present is defined as $|v_j| > 10^{-9}$.

Benjamini-Hochberg FDR correction is applied separately to each test column. A reaction is called significant if either test passes FDR < 0.05.

2.6 Biomarker Backtracking

Significant reactions were mapped to encoding genes via Recon3D GPR rules (Entrez IDs in format `G_<EntrezID>_AT1` resolved to HGNC symbols through MyGene.info) and to substrate/product metabolites via SBML stoichiometry.

2.7 Machine-Learning Pipeline

For each therapy, a per-patient feature vector was constructed from:
(i) *Metabolic features*: binary presence/absence of each biomarker reaction (47 for Chemo, 95 for Radiation). Binarisation removes scaling artefacts from cohort heterogeneity while preserving the GIMME on/off signal and it also behaves as a less stringent cutoff for prediction.
(ii) *Clinical features*: age at diagnosis, gender, race, ethnicity, disease type, primary site, AJCC pathologic stage, tumour grade, prior malignancy, and prior treatment. Categorical missing values carry an explicit `Unknown` level; numeric missingness is handled natively by LightGBM.

LightGBM [6] was selected for its handling of high-cardinality categoricals and missing values, strong tabular performance, and TreeSHAP compatibility. Hyperparameters were tuned with Optuna [1] (TPE sampler, 50 trials, 3-fold inner CV). Class imbalance was addressed with `scale_pos_weight` $= N_{\text{Resistant}}/N_{\text{Sensitive}}$. True performance was estimated by outer 5-fold stratified cross-validation. SHAP values were computed on the full re-trained model for evaluating the contribution of each feature in the model prediction.

3 Results and Discussion

3.1 Identified 47 Significant Chemotherapy Biomarkers in 1,719 Patients

Differential flux analysis identified 47 significant reactions at FDR < 0.05 (Table 2): 21 significant in both tests (*both*), 11 in the magnitude test only (*flux_only*), and 15 in the presence/absence test only (*presence_only*).

Table 2: Chemotherapy differential flux summary.

Metric	Value
Total reactions tested	10,600
Silent reactions	4,795
Active reactions	5,805
Significant reactions	**47**
Both	21
Flux only	11
Presence only	15

As shown in Figure 1, several of the top chemotherapy biomarkers are linked to known biological interpretations associated with resistance. The five most statistically significant biomarkers are listed in Table 3. In resistant patients, increased glutamate transport through *GLUt4* suggests enhanced antioxidant synthesis capacity via the TCA cycle, potentially helping tumour cells reduce reactive oxygen species (ROS) generated by chemotherapeutic drugs. Similarly, higher inositol transport through *INSTt2*, a precursor pathway for phosphatidylinositol lipid synthesis, may support PI3K/AKT/mTOR-mediated anti-apoptotic signalling and thereby promote cancer-cell survival under chemotherapy stress. Increased *NTD7* activity, involved in nucleotide salvage, may additionally contribute to chemoresistance because several nucleoside analogue drugs require phosphorylation for activation, whereas NTD7-mediated dephosphorylation can convert them back into inactive nucleosides. In contrast, reduced *RNDR1* flux in resistant tumours may indicate lower DNA repair activity following chemotherapy-induced DNA damage. Together, these biomarkers suggest that chemotherapy resistance is associated with coordinated alterations in antioxidant metabolism, survival signalling, nucleotide salvage, and DNA damage-response pathways.

Table 3: Top five Chemotherapy biomarkers.

Rxn	Pathway	$\log_2$FC	FDR
GLUt4	Glutamate transport	-1.82	1.1×10^{-9}
INSTt2	Inositol transport	-0.85	8.9×10^{-5}
H2OGLYAQPt	Aquaporin	$+0.34$	2.6×10^{-4}
NTD7	Nucleotide salvage	-1.07	2.6×10^{-4}
RNDR1	Ribonucleotide reductase	$+0.29$	5.1×10^{-7}

3.2 Identified 95 Significant Radiation Therapy Biomarkers in 1,595 Patients

The Radiation Therapy cohort yielded 95 significant reactions (Table 4), a roughly two-fold increase over Chemotherapy, reflecting the broader metabolic changes associated between sensitive and resistant cohorts in radiation therapy.

Table 4: Radiation Therapy differential flux summary.

Metric	Value
Total reactions tested	10,600
Silent reactions	5,006
Active reactions	5,594
Significant reactions	**95**
Both	48
Flux only	41
Presence only	6

As shown in Figure 2, several of the top radiation therapy biomarkers overlap strongly with the chemotherapy cohort, particularly *INSTt2* and *GLUt4*, suggesting the presence of a shared metabolic resistance signature across DNA-damaging therapies.The five most statistically significant biomarkers are listed in Table 5. Increased inositol transport through *INSTt2*, a precursor pathway for phosphatidylinositol lipid synthesis, may support PI3K/AKT/mTOR-mediated survival signalling and thereby enhance resistance to radiation-induced stress. Similarly, elevated glutamate transport through *GLUt4* may contribute to increased antioxidant synthesis via the TCA cycle, helping resistant tumours suppress reactive oxygen species generated during radiation exposure. Reduced *RNDR1* flux in resistant patients is additionally consistent with the findings of Lewis & Kemp (2021)[7], where nucleotide-pool maintenance was associated with improved DNA-damage repair capacity following radiation treatment. Lower lipogenesis-associated flux through *ACLYSt* may further indicate impaired membrane repair and recovery after radiation-induced cellular damage. Finally, altered folate-dependent metabolism through *FDPte* suggests broader perturbations in one-carbon metabolism and nucleotide biosynthesis in resistant tumours. Together, these biomarkers indicate that radiation resistance is associated with changes in antioxidant defence, survival signalling, DNA repair, lipogenesis, and one-carbon metabolic pathways.

Table 5: Top five Radiation Therapy biomarkers.

Rxn	Pathway	$\log_2$FC	FDR
INSTt2	Inositol transport	-1.62	2.4×10^{-12}
GLUt4	Glutamate transport	-1.88	2.8×10^{-11}
ACLYSt	ATP-citrate lyase	$+0.47$	3.6×10^{-7}
RNDR1	Ribonucleotide red.	$+0.35$	1.5×10^{-6}
FDPte	Folate transport	-0.46	1.9×10^{-6}

3.3 Small Cohorts: A Negative Result

For Hormone Therapy, Surgery, Immunotherapy, and Targeted Therapy ($77 \leq N \leq 116$), the permutation + Fisher's pipeline returned zero significant reactions even at relaxed FDR < 0.20. With $\sim$3,000 active reactions per cohort the multiple-testing penalty is severe, and small samples cannot generate p-values low enough to survive it. So, any future discovery in these

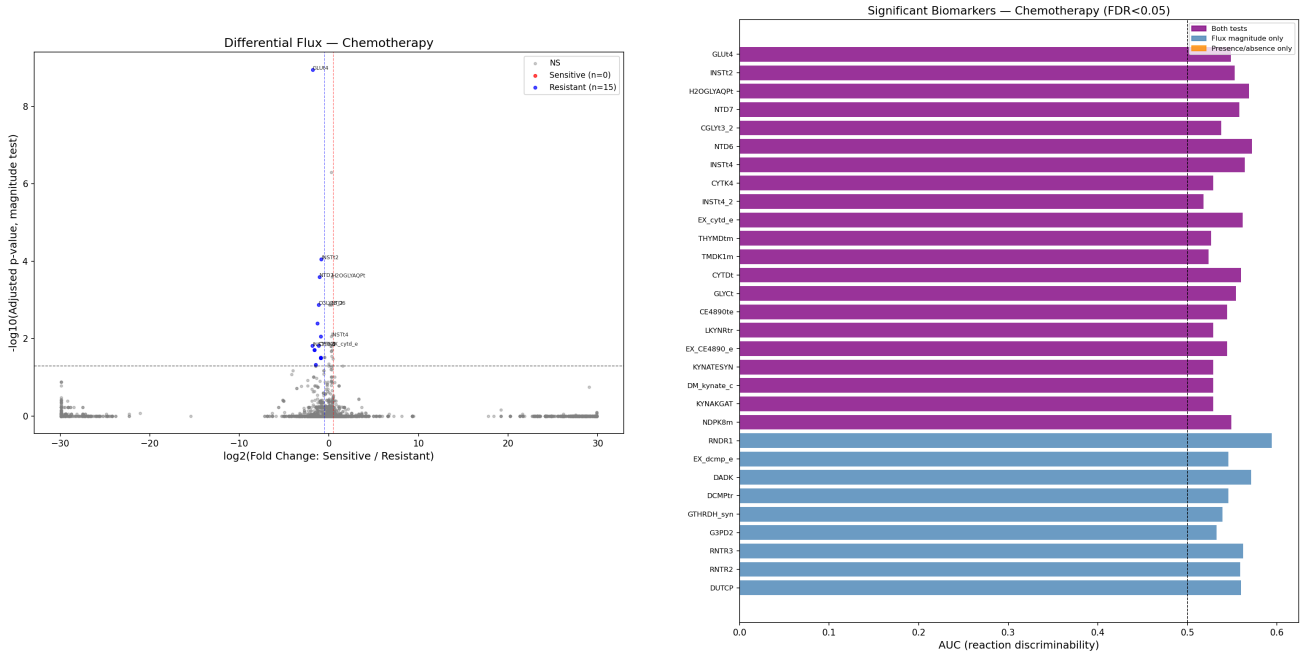


Figure 1: Chemotherapy: (left) volcano plot showing $-\log_{10}(\text{FDR})$ versus $\log_2$ fold change. (right) AUC ranking of significant biomarkers.

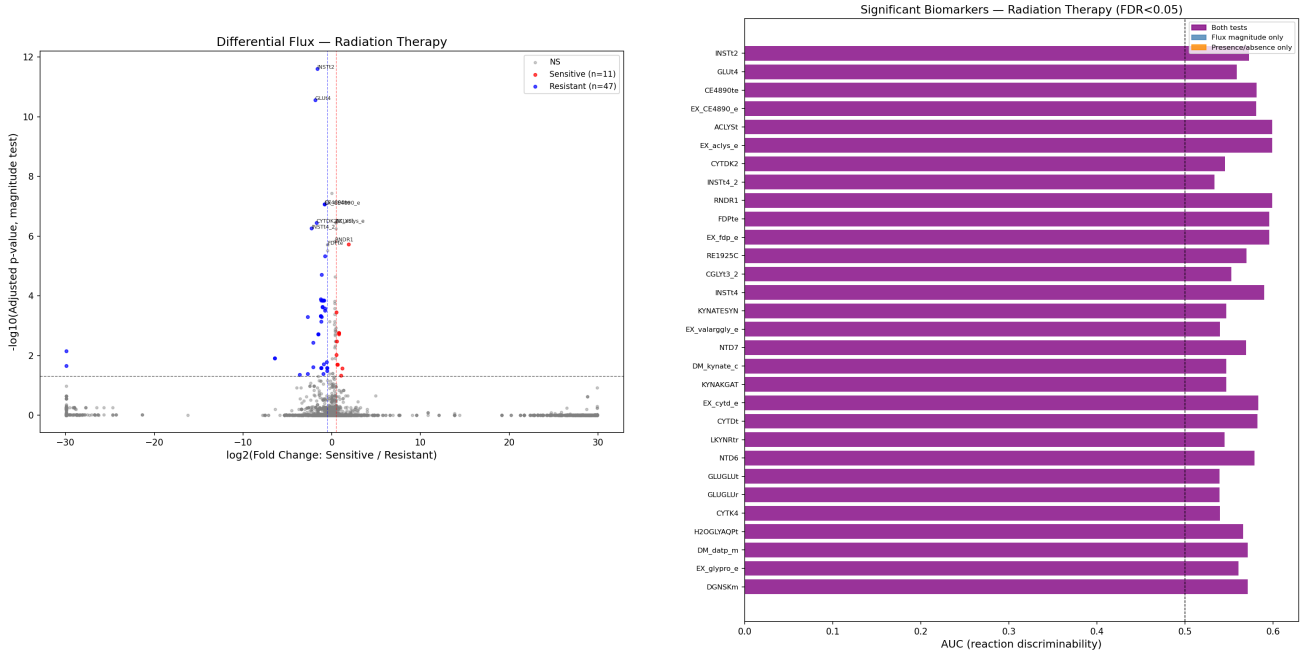


Figure 2: Radiation Therapy: (left) volcano plot showing $-\log_{10}(\text{FDR})$ versus $\log_2$ fold change. (right) AUC ranking of significant biomarkers.

therapies will require substantially larger cohorts, probably the controlled access data from TCGA could be of some help!

3.4 Reactions → Genes → Metabolites

GPR rules were decomposed into Entrez IDs and resolved to HGNC symbols via MyGene.info, yielding 53 unique genes for Chemotherapy and 81 for Radiation Therapy. For example, GLUt4 is encoded by *SLC38A3* (Entrez 10991) with substrates L-glutamate and sodium; INSTt2 is encoded by *SLC2A13* (Entrez 114134), transporting myo-inositol with a proton co-substrate. The top 5 biomarkers in each of the therapy category, not sorted by statistical significance this time, but by log2FC scores on significant biomarkers, are the canidataes for further wet - lab validation. The further steps would also include to take all the significant biomarkers obtained from both therapies for literature review to find out the novel biomarkers obtained in this study.

3.5 Radiation Therapy Response Prediction Achieved 0.82 OOF AUC Compared with 0.78 for Chemotherapy

Table 6 shows stratified 5-fold CV results. The Radiation model consistently performs better compared to the Chemo model (AUC 0.818 vs 0.785; σ 0.013 vs 0.020). The Chemo model's Fold-3 balanced accuracy of 0.500 reflects a fold where the threshold-0.5 classifier collapsed to a constant prediction — an expected problem due to class imbalance — which is why out-of-fold (OOF) aggregated scores are reported as the primary metric. Overall both the models perform well in predicting therapy efficacies for cancer patients for both therapies.

Table 6: Stratified 5-fold cross-validation results.

Therapy	Fold	AUC	BAC	F1
Chemo	1	0.769	0.677	0.764
	2	0.814	0.743	0.801
	3	0.826	0.500	0.791
	4	0.817	0.754	0.778
	5	0.799	0.733	0.777
	OOF	**0.785**	**0.681**	**0.783**
Radiation	1	0.845	0.740	0.833
	2	0.831	0.743	0.858
	3	0.817	0.760	0.875
	4	0.806	0.732	0.837
	5	0.820	0.752	0.830
	OOF	**0.818**	**0.745**	**0.847**

3.6 Biomarker Reactions Emerge as Interpretable Secondary Predictors

As shown in Figures 3 and 4, the dominant SHAP features for both therapies are primarily clinical variables, including tissue or organ of origin, primary cancer site, AJCC stage, and age at diagnosis. This is biologically expected, since cancer type and disease stage are among the strongest predictors of treatment response, and a predictive model excluding these variables would be meaningless. Importantly, several metabolic reaction features additionally appear among the top-ranked SHAP variables for both therapy classes. For Chemotherapy, reactions such as *rxn_NDPK8m*, *rxn_GLYCt*, *rxn_DM_datp_m*, *rxn_PUNP3*, and *rxn_G3PD2* contributed measurable predictive signal, whereas Radiation Therapy prominently involved reactions including *rxn_EX_HC00342_e*, *rxn_GLUPROASCT1*, *rxn_H2O2syn*, and *rxn_CYTK2*. The SHAP beeswarm distributions further show that the directionality of these metabolic features is broadly consistent with the differential flux analysis presented earlier, supporting the biological coherence of the model outputs. Together, these findings suggest that GSMM-derived metabolic features function as interpretable secondary predictors: they do not replace core clinical variables, but instead explain an additional component of therapy-response variability in a meaningful manner. From this analysis, as well we can take the top performing biomarkers with respect to the SHAP values and further wet-lab validation may be done.

3.7 Biomarker Reactions Show Consistent Directionality Across Chemotherapy and Radiation Therapy

One of the most interesting features of Figure 5 is the absence of any data point in the off-diagonal quadrants among the 37 shared biomarkers (purple circles). Every reaction that is elevated in chemo-resistant patients is also elevated in radio-resistant patients, and conversely, every reaction enriched in chemo-sensitive patients is likewise enriched in radio-sensitive patients. Not a single reaction flips direction between the two axes. This means there is no reaction that is higher in chemo-sensitive but radio-resistant tumours, or vice versa — the upper-left and lower-right quadrants are completely empty of shared biomarker points. This implies that the therapy resistance in these cancer patients is driven by a shared underlying metabolic dysfunction, not by mechanisms that are specific to how each modality kills cells. But, we do have reactions that are unique to each category that serve as the markers for prediction. The purple cluster in Quadrant I is dominated by reactions of nucleotide metabolism. Reactions such as `RNDR1` (ribonucleotide reductase) and `NDPK8m` (nucleoside-diphosphate kinase, mitochondrial isoform) are elevated in therapy-sensitive patients across both treatment axes. an elevated rate of nucleotide cycling implies that the tumour's replication machinery is actively engaged. Cells in active S-phase are considerably more vulnerable to both DNA-damaging chemotherapy agents and radiation, because the lesions inflicted are more likely to be encountered by a replication fork before they are repaired, converting sub-lethal damage into lethal double-strand breaks. The chemo-only reactions largely centre around thymidine salvage, mitochondrial nucleoside phosphorylation, and organic acid transport — pathways that are particularly relevant when cytotoxic drugs target nucleotide synthesis or DNA repair machinery directly. Few reactions from the plot, that have high —log2FC— values, indicating signficant difference in expression levels are:

1. **HC00900t2** – Transport of Methylmalonate. This reaction carries the **largest magnitude of change** in the entire dataset, with a $\log_2$FC of approximately -29.9, higher in radio-resistant. It mediates the antiport of methylmalonate, a branched-chain organic acid, against sulphate across the mitochondrial or plasma membrane. Elevated methylmalonate antiport in radio-resistant patients suggests increased propionate and odd-chain fatty-acid catabolism, potentially supplying alternative carbon sources to maintain TCA-cycle activity and ATP production under radiation stress. Altered sulphate transport may additionally influence antioxidant capacity through glutathione-associated sulphur metabolism, a known contributor to radioresistance.
2. GLYCLTt – Glycolate Transport via Diffusion. It

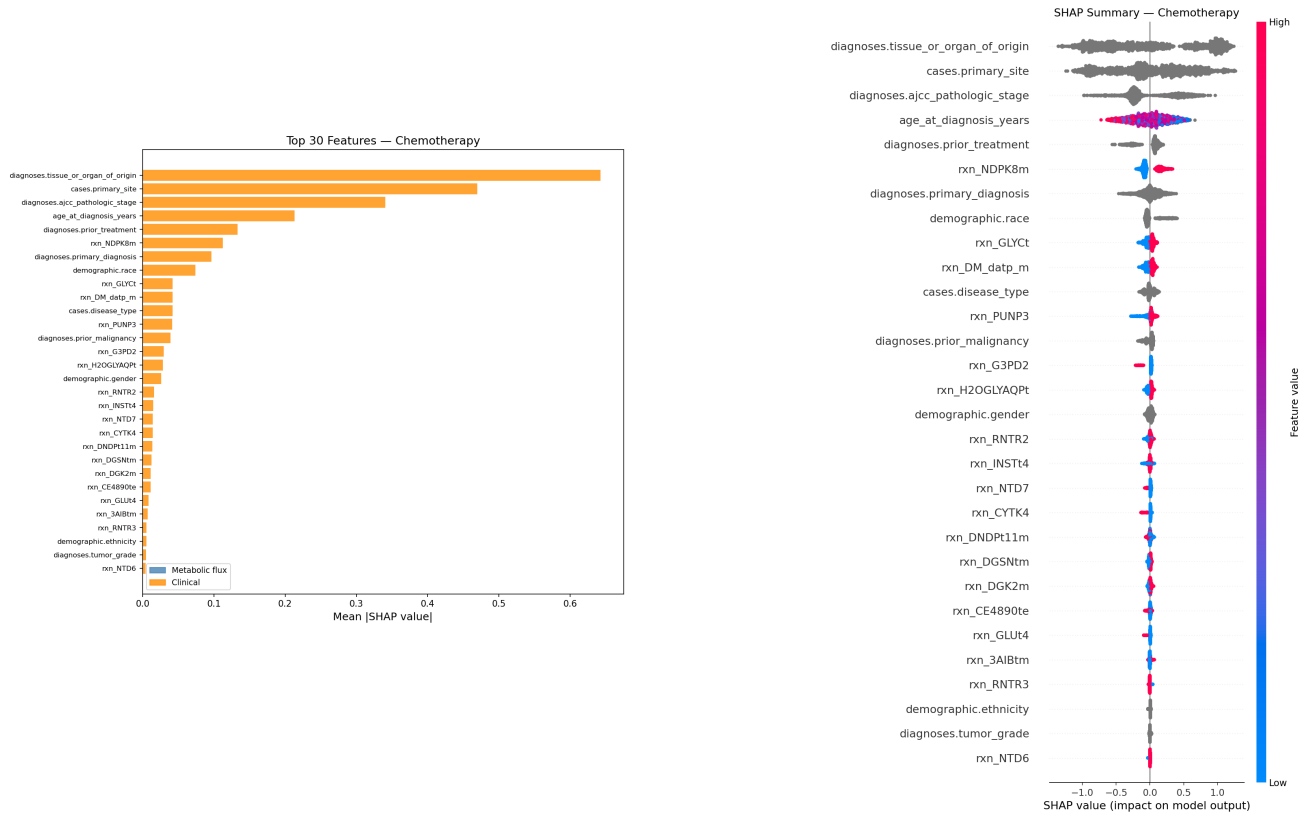


Figure 3: Chemotherapy SHAP analysis. (left) Top 30 features ranked by mean absolute SHAP value. (right) SHAP beeswarm plot showing per-patient feature impact and feature-value direction (red = high feature value, blue = low).

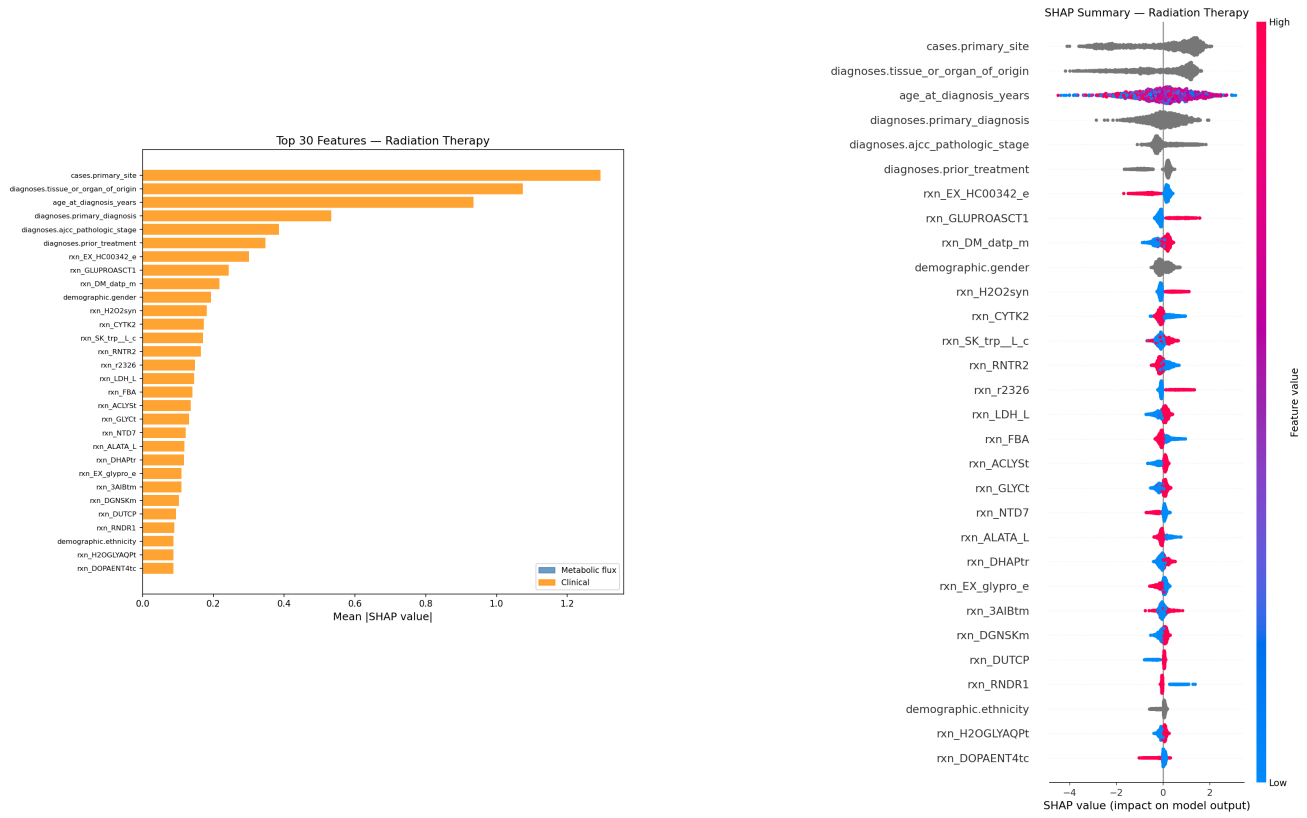


Figure 4: Radiation Therapy SHAP analysis. (left) Top 30 features ranked by mean absolute SHAP value. (right) SHAP beeswarm plot showing per-patient feature impact and feature-value direction (red = high feature value, blue = low).

showed the strongest chemotherapy-associated shift, with substantially higher flux in chemo-resistant

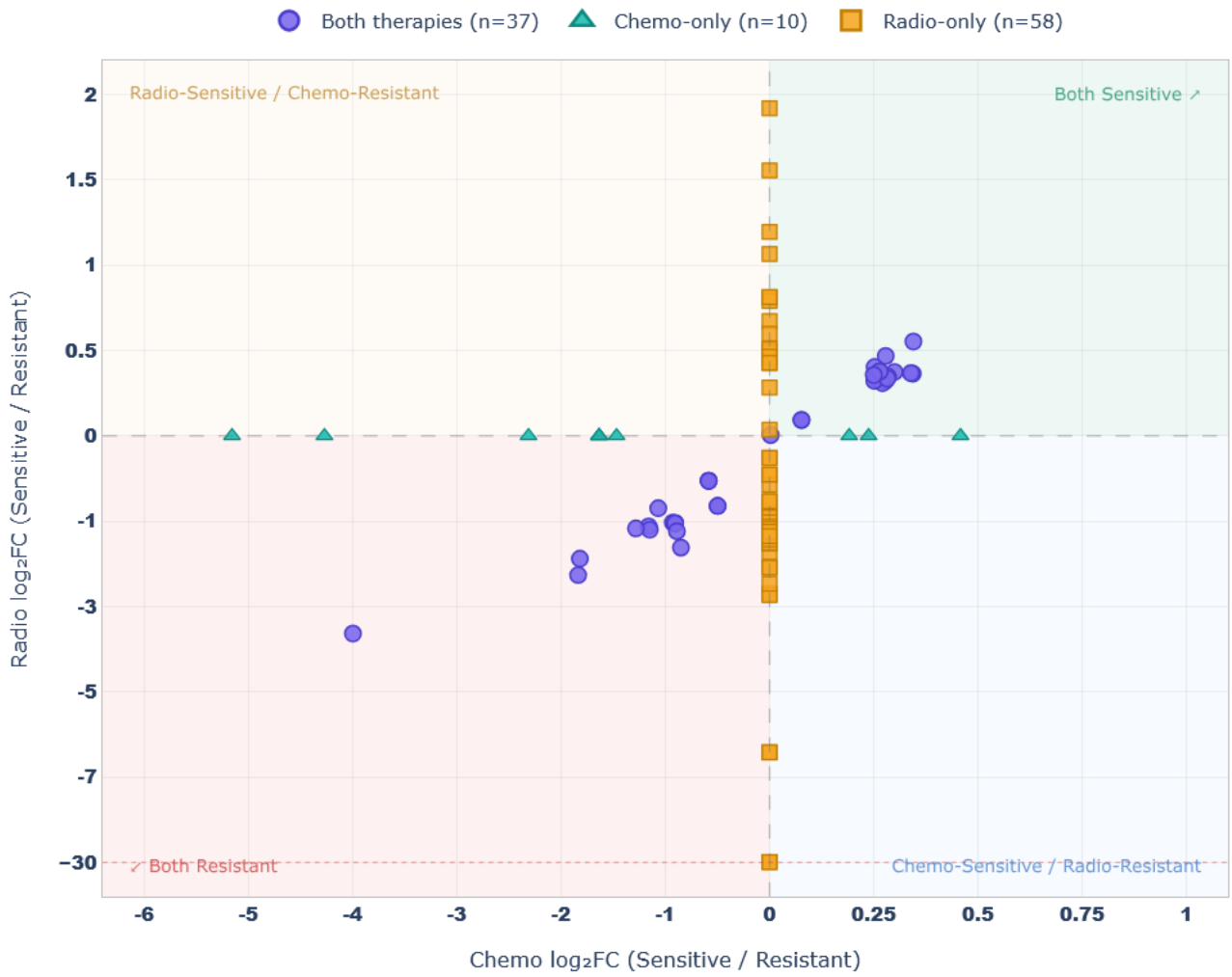


Figure 5: Dual-therapy metabolic biomarker scatter plot. Each point represents a statistically significant reaction flux difference between sensitive and resistant patient groups. The x-axis shows the $\log_2$ FC for chemotherapy (Sensitive/Resistant), and the y-axis shows the corresponding value for radiotherapy. Points are stratified by therapy overlap: **purple circles** represent reactions significant in both therapies; **teal triangles** represent chemo-only biomarkers; and **orange squares** represent radio-only biomarkers. Quadrant backgrounds indicate the expected resistance phenotype: upper-right (both sensitive), lower-left (both resistant), upper-left (radio-sensitive/chemo-resistant), and lower-right (chemo-sensitive/radio-resistant). Interactive HTML visualisation for this plot is given in the following html file.

patients ($\log_2$FC = −5.16). Increased glycolate transport suggests elevated glyoxylate and one-carbon metabolism activity, which may help resistant tumours compensate for folate-pathway inhibition caused by drugs such as methotrexate and 5-fluorouracil. This reaction is therefore likely not a direct resistance mechanism itself, but instead reflects a broader metabolic adaptation supporting nucleotide synthesis and survival under chemotherapy stress.

3. DHAPtr – Transport of Dihydroxyacetone Phosphate. It was strongly enriched in radio-sensitive patients, indicating higher transport of the glycolytic intermediate dihydroxyacetone phosphate (DHAP) with $\log_2$FC = +1.92. Increased DHAP flux may enhance glycerophospholipid metabolism and mitochondrial redox shuttling, both of which can amplify reactive oxygen species production during radiation exposure. This could make sensitive tumours more vulnerable to lipid peroxidation and oxidative damage, consistent with the broader observation that radiation-sensitive tumours exhibit metabolic states that favour damage amplification rather than resistance.

3.8 Comparison with Lewis & Kemp (2021)

Table 7 places this work alongside the most directly comparable prior study. The principal contributions are cohort scale (12.5×), multi-therapy stratification, reaction-level resolution and comparison of biomarkers between therapies to reach the ultimate goal of therapy efficacy prediction, a completely different objective. AUC is comparable (0.818 vs ≈ 0.85); the present cohort is far better powered except for the missing the inclusion of other multi-omics datasets other than the transcriptome, and the framework generalises beyond a single therapy. Lewis and Kemp's[7] 236 metabolite-feature

count is not directly comparable: the 47 + 95 reactions identified here trace back to 53 + 81 unique genes and dozens of substrate/product metabolites which, if recast in their metabolite-level framing, would yield a comparable or larger number. This study also lacks the wet lab validation done by the Lewis and Kemp's [7] study.

Table 7: Comparison with Lewis & Kemp (2021)[7].

Dimension	Lewis & Kemp 2021	This work
Patients	915	11,505
Therapies	Radiation only	6 (2 labelled)
Feature type	Metabolite rates	Rxns + clinical
Statistical tests	Mann-Whitney + Fisher's	Mann-Whitney + Fisher's + Permutation
Biomarkers	236 metabolites	47 chemo + 95 radiation rxns
ML algorithm	GBM + SHAP	LightGBM + Optuna + SHAP
Best AUC	≈ 0.85	0.82 (radiation)

4 Conclusion

This work demonstrates that patient-specific GSMMs built at the scale of 11,505 TCGA patients can identify metabolic biomarker reactions of cancer therapy response in a statistically rigorous and biologically interpretable way. The GIMME-based pipeline recovers 47 Chemotherapy and 95 Radiation Therapy biomarker reactions, traces them to 53 and 81 unique genes respectively, and supports per-therapy LightGBM classifiers reaching OOF AUCs of 0.785 and 0.818 with full SHAP interpretability.

The important finding is that clinical features (cancer type, AJCC stage) dominate prediction, with metabolic features providing mechanistically meaningful secondary signal dominating over some of the other clinical features as well. Together, these biomarker reactions indicate that therapy resistance is not driven by a single metabolic pathway, but rather by coordinated rewiring of antioxidant defence, nucleotide metabolism, lipid remodelling, amino-acid catabolism, and survival signalling pathways. Notably, the extreme radio-resistant enrichment of `HC00900t2` ($\log_2$FC ≈ -30) highlights how certain transport reactions may become nearly exclusive metabolic dependencies in resistant tumours.

5 Future Directions

1. Apply trained models to ˜7,800 unlabelled TCGA patients and output a per-patient therapy ranking (Chemotherapy versus Radiation Therapy) with confidence intervals.
2. Inclusion of controlled-access TCGA transcriptomic data could potentially recover biomarkers for the remaining therapy classes that were excluded due to insufficient cohort size.
3. Perform wet-lab validation for the top biomarker reactions using knockdown experiments in cell lines and compare resistance phenotypes against model predictions.
4. Construct cell-type-specific GSMMs from available TCGA Pan-Cancer single-cell RNA-seq datasets to separate tumour-intrinsic metabolism from microenvironment-associated metabolic effects.
5. Develop a pan-therapy multi-task learning framework capable of jointly predicting sensitivity across all therapy classes, allowing identification of cross-therapy metabolic resistance signatures currently lost in per-therapy models.

6 GitHub Link

All code files used can be found in this repository.

7 Progress After Intermediate Results Presentation

At the time of the intermediate results presentation, data acquisition and pipeline integration were the primary challenges — bulk-downloading 11,505 RNA-seq files from the GDC portal and establishing a reproducible HPC workflow consumed the bulk of effort. The intermediate slides showed only basic sample-distribution charts and a preliminary sensitivity-vs-resistance bar plot.

After that checkpoint the entire pipeline was completed:

- **GSMMs construction**: all 11,505 GIMME models were built and stored as COBRApy pickles.
- **FBA matrix**: the 11,505 × 10,600 flux matrix was assembled and QC-validated.
- **Differential flux analysis**: both magnitude (Mann-Whitney/permutation) and presence/absence (Fisher's) tests were implemented, with BH-FDR correction and the four-class biomarker taxonomy.
- **Backtracking**: GPR-rule decomposition and MyGene.info resolution yielded gene and metabolite annotations for all significant reactions.
- **ML pipeline**: LightGBM classifiers with Optuna tuning and SHAP interpretability were trained and cross-validated for both Chemotherapy and Radiation Therapy.

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